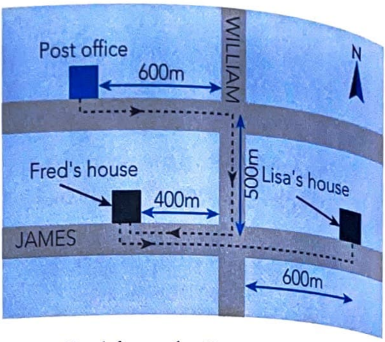
REVIEW QUESTIONS

Chapter 4: Linear Motion and Force

1. Differentiate between:
   (a) distance and displacement;
   (b) speed and velocity.

2. Lisa collects a parcel from the Post Office, walks 600 m East then turns South into William Street. On reaching James Street she walks 400 m West where she delivers the parcel at Fred's house. Lisa then walks East along James Street towards her house as shown.



   (a) Calculate the total distance that Lisa walked from the Post Office to her house.

   (b) Calculate her total displacement (magnitude only) from the Post Office to her house.

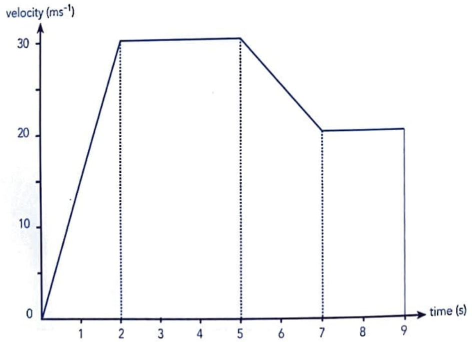
3. During a physics experiment to determine the average velocity of a dynamics trolley, Sally obtained the following data:

TIME (s)	0	1	2	3	4	5
DISPLACEMENT (cm)	0	0.8	1.6	2.4	3.2	4.0

   (a) Draw a displacement-time graph using this data.

   (b) Use the graph to determine the average velocity of the trolley.

4. A vehicle started from rest at t = 0 and continued moving in a straight line for 9 s. An analysis of its motion is represented in the graph below:



   (a) Briefly describe the vehicle's motion throughout the 9 s time interval.

   (b) Calculate the vehicle's displacement after 9 s.

   (c) Compare the vehicle's acceleration for the first 2 s with that for the interval between the 5th and 7th second.

5. When Andrew drove his car around a tight bend on a narrow mountain road a 5 kg piece of luggage was dislodged from the car's roof rack. The piece of luggage just rolled beneath the road's safety barrier and fell vertically down a ravine. On returning to the point where the piece of luggage fell into the ravine Andrew noticed that a 1 kg stone took 5 s from rest to reach the bottom of the ravine. Calculate the depth of the ravine.

6. Janice, standing on the edge of a traffic bridge which is 10 m above a river, throws a small stone 20 m vertically into the air. Calculate the:
   (a) initial velocity with which the stone left her hand, and
   (b) the time taken for the stone to reach the water (ignore any effects due to friction).

7. Explain the difference between a scalar quantity and a vector quantity.

8. Classify the following quantities as either a scalar quantity or a vector quantity.

   distance, force, mass, displacement, acceleration, velocity, volume, time

9. A young child is restrained in a shopping trolley at a supermarket when the child's older brother pushes the trolley from rest with a force of 15 N. If the child's older sister simultaneously applies a force of 10 N on the trolley in the same direction, calculate the total force received by the trolley and child.

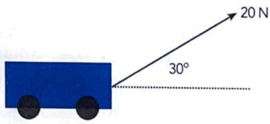
10. If the trolley and young child in question 9 received a retarding force of 3 N when the child's older brother pushes the trolley from rest with a force of 15 N, calculate the total force received by the trolley and child.

11. A car being driven North at 12 ms^{-1} rounds a corner so that it is now being driven West at 16 ms^{-1}. Calculate the change in velocity of the car.

12. John is a passenger on a train which has a velocity of 60 kmh^{-1}. Calculate John's velocity, relative to the ground if he walks towards the
   (a) front of the train at 2 kmh^{-1},
   (b) back of the train at 2 kmh^{-1}.

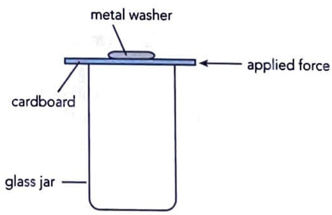
13. An athlete was running due East at 5 ms^{-1} when a southwesterly wind blowing at 2 ms^{-1} changes both his speed and the direction in which he was running. Calculate the magnitude of his new velocity.

14. A force of 20 N is applied to a small cart at an angle of 30° above the horizontal. Calculate the horizontal and vertical components of the force.



15. List four effects produced by forces.

16. Mark places a metal washer on a piece of cardboard which rests on top of a glass jar as shown in the diagram.



   (a) Predict what will happen to the washer when Mark quickly flicks the cardboard with his finger.
   (b) Explain your prediction.

17. State Newton's First Law of Motion.

18. When driving her car at a constant velocity of 60 kmh^{-1} on a flat road Mary found it necessary to keep her foot on the car's accelerator. Is Newton's First Law of Motion being broken? Why?

19. Nic was sitting in the dining car of the Indian Pacific train pouring himself a cool drink from a bottle into a glass standing on the table. As he was doing this the train braked quickly. Nic was quite proud that he was still able to hold the mouth of the bottle above the glass (which did not fall over) as the train braked.
    (a) Why was Nic proud of this achievement?
    (b) Would you expect all of the cool drink to pour into the glass during the braking?

20. State Newton's Second Law of Motion.

21. When observing the world land speed record for a vehicle a journalist was heard to say "the rapid increase in acceleration could cause the driver's internal stomach organs to receive enough force to produce severe stomach pains". Is the physics in this statement strictly correct? Explain your answer.

22. Kim was driving his car home after doing some shopping when he missed the turn-off to his street. Without thinking he braked hard causing his car to stop quickly. He looked into the back of the car to find that an egg carton, full of eggs, that he just bought at the shop had dislodged itself from the back seat onto the floor. On inspecting the carton he was surprised to find that not one egg was broken.

    Explain why:
    (a) the egg carton fell to the floor during braking, and
    (b) the material used in the egg carton prevented the eggs from breaking.

23. State Newton's Third Law of Motion.

24. Jacky and Susan were keen to enter the running races in the school sports. Neither had used starting blocks before and both decided to try them during a practice session. Susan attached her starting blocks to the ground with a peg while Jacky did not attach her starting blocks to the ground. Which runner is most likely to have the best start from their starting blocks? Explain your answer.

25. Tim played squash in hard, flat soled shoes and after most games he suffered from sore feet and legs. When he switched to wearing soft, thick rubber soled shoes while playing squash, he no longer got sore feet and legs. Use Newton's Second Law of Motion to explain why the soft, thick rubber soled shoes did not cause sore feet and legs.

26. A journalist writing about the safety features of a recently released car wrote "the purpose of the air bag is to reduce injury to the driver if the car is involved in a collision". Briefly explain how the air bag can reduce injury to the driver.

27. A 40 kg ice skater moving at 3 ms^{-1} on wet ice collides with an ice skater of mass 30 kg moving in the opposite direction at 5 ms^{-1}. If the skaters hold hands and continue moving in the same straight line after the collision, calculate their common velocity.

28. Peta, a keen softball player, was having difficulty hitting the ball hard and therefore past nearby fielders. She was advised that her problem was related to her failure to 'follow through' when striking the ball. Explain why to 'follow through' is important in Peta's case.

29. Calculate the force on Vic's hand if he catches a 0.1 kg ball moving at 10 ms^{-1}. It is estimated that the ball came to rest 0.1 s after hitting Vic's hand.

30. In question 10 a shopping trolley with a young child in it is being pushed from rest with a force of 15 N against a retarding force of 3 N. This force is applied for 4 s. If the trolley and the young child have a combined mass of 60 kg, calculate
   (a) the acceleration of the trolley and child,
   (b) the change in momentum of the trolley and child, and
   (c) the distance travelled by the trolley and child during the time the force was applied.

31. Define the terms (a) mass, and (b) inertia.

32. A crane lifts a 1500 kg car 3.0 m vertically to place it onto the tray of a transport carrier truck. Calculate the:
   (a) force exerted, and
   (b) work done by the crane.

33. Theresa was driving her car of total mass 1500 kg at 60 kmh^{-1} on a level road when she released her foot from the accelerator, put the gear into neutral and 'free-wheeled' along the road until the car stopped (Not a wise thing to do!). Calculate the:
   (a) initial kinetic energy of the car,
   (b) stopping force (assume to be constant) if the car travelled 120 m before stopping,
   (c) time taken to stop, and
   (d) work done in stopping the car.

34. A small truck of mass 8000 kg is being passed on the freeway by a car of mass 1600 kg. If at the point of passing the car is travelling at 100 kmh^{-1} and has the same kinetic energy as the truck, what is the truck's velocity?

35. A toy car of mass 100 g is powered by a spring mechanism. If the energy transfer is 100% efficient and the potential energy of the spring at the moment the car is released is 2 J, calculate the speed with which the car can travel.

36. Jane threw a dart of mass 50 g into a block of soft wood. The block of mass 1.0 kg, was suspended freely by a length of fishing line before being hit by the dart. Andrew observed that the block with the dart in it swung to one side and rose a vertical distance of 40 cm. Calculate the speed of the dart when it hit the block.

37. Alan had difficulty pushing a peg of mass 200 g which was used to hold a rope to his tent into the ground so he decided to jump onto the peg. He jumped to a height of 30 cm and landed vertically on top of the peg which was 15 cm above the ground. To his dismay the peg only moved 9.0 cm into the ground.

   If Alan has a mass of 45 kg, find:
   (a) his energy just before impact with the peg,
   (b) his velocity just before impact with the peg,

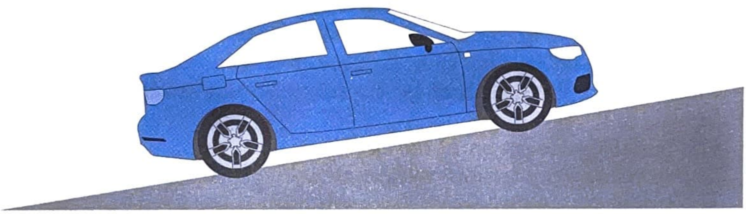
(c) the velocity of the peg after impact if Alan's velocity after the impact is the same as that of the peg,
(d) the energy loss due to the impact,
(e) the average force of friction due to the ground which acted on the peg.

38. During a physics experiment to measure the power necessary to run up a flight of stairs, Maree recorded a time of 10 s for Shirley to run up the stairs. The two girls also recorded the following data:

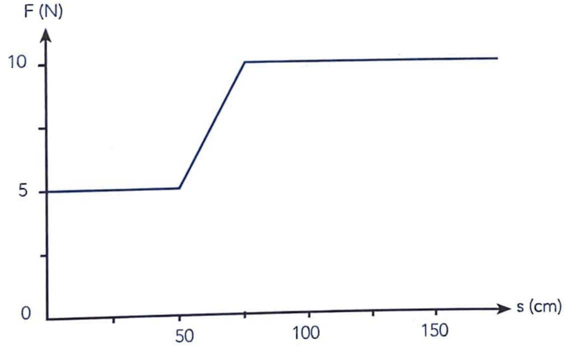
Shirley's mass	= 50 kg
number of steps climbed	= 30
height of each step	= 25 cm

Calculate:
(a) the work done by Shirley, and
(b) her power output on reaching the top of the stairs.

39. Sam drives his car of mass 1500 kg up a 10° hill. The force due to gravity along such an incline is given by m × g × sin 10°. Given that the total resistance due to friction is 200 N and that he is driving the car at a constant speed of 15 ms^{-1}, calculate the power developed by his car's engine.



40. In a physics experiment Kim applied a force of 5.0 N to a wooden block over a distance 50 cm then increased the force to 10 N. She recorded data for the applied force and the block's displacement to plot the following graph.



Calculate the:
(a) work done in moving the block 150 cm,
(b) average power exerted by Kim if she took 10 s to displace the block 150 cm.